Diversity Measures and Ranking Procedures in PODEX

Niloufar Abhari

Fall 2025

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1 Overview

This document explains the diversity measures and ranking procedures implemented in `diversity.py`. The goal of these methods is to quantify how much genetic diversity is expected to be maintained under population extinction risk, and to rank populations according to their importance (contributing more diversity) for conservation.

Throughout, we assume:

- n populations indexed by $i = 1, \ldots, n$
- m loci indexed by $j = 1, \ldots, m$
- p_{ij} = allele frequency (or presence/absence) of population i at locus j
- N_i = population size
- e_i = extinction probability of population i

The input frequency matrix is therefore $n \times m$.

2 Pooled Heterozygosity (Het)

2.1 Deterministic pooled heterozygosity

Function: `Het_Pooling_with_pop_size`

Definition. For each locus j, the size-weighted pooled allele frequency is

$$\bar{p}_j = \frac{\sum_{i=1}^{n} N_i p_{ij}}{\sum_{i=1}^{n} N_i}.$$

The pooled heterozygosity is

$$H_{\text{pool}} = \sum_{j=1}^{m} 2\bar{p}_j(1 - \bar{p}_j).$$

Intuition. This measures how genetically variable the pooled population would be if *all* populations survive and are combined, weighting each population by its size.

Small example. Two populations, one locus:

$$p = (0, 1), \quad N = (50, 50).$$

Then $\bar{p} = 0.5$ and $H_{\text{pool}} = 2(0.5)(0.5) = 0.5$.

2.2 Monte Carlo pooled heterozygosity with extinction

Function: `Het_Pooling_mc_with_extinction`

Definition. Each population survives independently with probability $1 - e_i$. For each Monte Carlo trial:

1. draw a survival vector $S_i \sim \text{Bernoulli}(1 - e_i)$,
2. compute $\bar{p}_j$ using only surviving populations,
3. compute H_{pool} for that trial.

The output is the average of H_{pool} over all trials.

Intuition. This gives the *expected* pooled heterozygosity when populations may go extinct. Trials in which many populations go extinct typically reduce diversity.

Key difference from deterministic Het. Deterministic Het assumes survival is guaranteed. Monte Carlo Het explicitly averages over extinction outcomes.

3 Split System Diversity (SSD)

3.1 Deterministic SSD pooling

Function: `SSD_Pooling_with_pop_size`

Definition. For each locus j, compute the weighted mean frequency $\bar{p}_j$ as above. A locus contributes 1 if

$$0 < \bar{p}_j < 1,$$

and 0 otherwise. SSD pooling is the total number of polymorphic loci.

Intuition. SSD measures *how many loci remain polymorphic* in the pooled population, rather than how polymorphic they are.

Example. If 10 loci exist in the pooled metapopulation and 7 have $0 < \bar{p}_j < 1$, then SSD pooling $= 7$.

3.2 Monte Carlo SSD pooling with extinction

Function: `SSD_Pooling_mc_with_extinction`

Definition. As with Monte Carlo Het, survival is sampled repeatedly. For each trial, the number of polymorphic loci is counted. The output is the mean number of polymorphic loci across trials.

Intuition. This is the expected number of loci that remain polymorphic after accounting for extinction.

4 Shapley Values

4.1 Shapley values without extinction

Functions: `shapley_values_exact`, `shapley_values_mc`

Definition. The Shapley value of population i is its average marginal contribution to diversity, averaged over all possible orders in which populations could be added:

$$\phi_i = \mathbb{E}_\pi\big[v(S_i^\pi \cup \{i\}) - v(S_i^\pi)\big],$$

where $v(\cdot)$ is a diversity score and S_i^π is the set of populations appearing before i in permutation π.

Intuition. Shapley values fairly allocate total diversity among populations.

4.2 Shapley values with extinction

Function: `shapley_values_mc_with_extinction`

Definition. For each Monte Carlo iteration:

1. sample which populations survive,
2. compute Shapley contributions only among survivors,
3. accumulate marginal contributions.

Expected vs conditional Shapley values.

- **Expected (return_conditional=False):** averages contributions over all trials, including trials where a population is extinct (contribution = 0). So if a population often goes extinct (e.g. 90% chance), its expected contribution will be small because in most Monte Carlo runs it's not even there to add diversity.
- **Conditional (return_conditional=True):** averages contributions only over trials where the population survives.

5 Expected Diversity Gain $W(i)$ and iHED-style Ranking

5.1 Baseline gain $W(i)$

Definition. Let $H(\mathbf{e})$ be expected diversity under extinction probabilities $\mathbf{e}$. The baseline gain of population i is

$$W(i) = H(\mathbf{e}^{(i=0)}) - H(\mathbf{e}),$$

where $\mathbf{e}^{(i=0)}$ sets $e_i = 0$ and leaves all other e_j unchanged.

Intuition. Baseline $W(i)$ answers: *If I could save exactly one population, which one gives the biggest expected gain?*

5.2 Iterative (iHED-style) ranking

Definition. Populations are protected sequentially:

1. compute baseline gains,
2. protect the best population (set its $e_i = 0$),
3. recompute gains for the remaining populations,
4. repeat.

Intuition. This mimics a conservation strategy where protection happens step-by-step and earlier choices affect later ones.

5.3 Are the top k $(k < n)$ populations in iHED the same as "protect only k populations"?

No, not necessarily. Protecting exactly k populations without iteration corresponds to selecting the top k *baseline* gains. iHED-style ranking recomputes gains after each protection. Because diversity is nonlinear, protecting one population can change the marginal value of others. Therefore, the top k baseline populations may differ from the first n populations chosen by the iterative procedure (see [1]).

5.4 How different are baseline gains and iterative gains?

Baseline gains measure *individual importance in isolation*. Iterative gains measure *additional importance given previous protections*.

They differ because protecting populations stabilizes allele frequencies by preventing them from collapsing to 0 or 1, redundancy among populations increases, marginal gains typically decrease but not monotonically. Therefore, baseline and iterative rankings answer different conservation questions and should not be expected to coincide.

References

[1] Niloufar Abhari et al. "Measuring genetic diversity across populations". In: *PLOS Computational Biology* 20.12 (2024), e1012651.